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19 October 1964
SUPERSEDING
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5 July 1962
MIL-M-1250 (in part)
27 July 1949

MILITARY SPECIFICATION

FORMING MACHINES, SHEET METAL, BAR FOLDING MACHINE, AND GROOVING MACHINES, HAND OPERATED, BENCH TYPE

This specification is mandatory for use by all Departments and Agencies of Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers hand operated bench type sheet metal forming machines of the bar folding and grooving types, with equipment and accessories as described herein.

1.2 Types.- Sheet metal forming machines covered by this specification shall be hand operated for bench mounting of the following types as specified (see 6.2):

Type I - Bar folding
Type II - Grooving

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-18058 - Preparation for Delivery of Machine Tools (Metals and Woodworking), Accessories and Repair Parts.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards
Handbook H28 Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, 25, D. C.)

3. REQUIREMENTS

3.1 Design.- The sheet metal machines furnished under this specification shall be new, the manufacturer's latest design which have proved successful in operation and dependable in service.

3.1.1 Lubrication.- Bearings, revolving shafts and moving parts shall be provided with ample and suitable means of lubrication, as required.

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3.1.2 Interchangeability.- All parts shall be manufactured to such standards as will permit replacement or adjustment without modification of parts or machine.

3.1.3 Material.- All parts used in the fabrication of the machine shall be new and of the quality necessary to produce a machine to meet the requirements described herein.

3.2 Construction.- The machine shall be complete so that when installed it can be used for any operation for which it was designed. Construction shall be free from any characteristics or defects that will prevent the machine from passing any of the inspections in 4.2.

3.2.1 Castings and forgings.- All castings and forgings shall be free from defects, scale, and mismatching. In no event shall such processes as welding, peening, plugging, or filling with cold solders or metallic pastes be used on castings or forgings for reclaiming any parts of the machine without authorization.

3.2.2 Welding, brazing or soldering.- Welding, brazing or soldering shall be employed only where those operations are included in fabrication of the original design. These operations shall not be employed as repair measures for defective parts.

3.2.3 Fastening devices.- All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.2.4 Surfaces.- All surfaces of all castings, forgings, molded parts, stampings, and welded parts shall be cleaned and free from sand, dirt, fins, sprues, scale, flux or other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

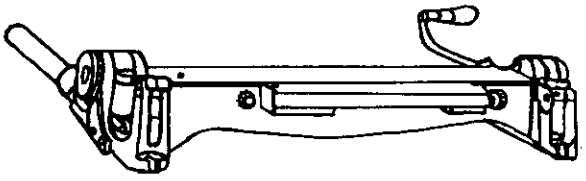
3.2.5 Painting.- Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.2.6 Threads.- All threaded parts shall be in the inch system and shall conform to H28.

3.3 Detailed requirements.-

3.3.1 Type I forming machine.- Type I forming machine shall be of the adjustable bench type, hand lever operated and shall be suitable for making folds of limited widths on the edges of sheet metal to form locks, square joints or turn edges for inserting wire. Type I machines shall consist of a base, frame, folding blade, gage stops, clamping jaw and the necessary accessories and operating levers.

3.3.1.1 Base frame.- The frame shall be of rugged construction and shall be of alloy steel or gray iron casting of such size and weight to provide support for the folding bar, gages, stops adjusting devices, operating levers and other attachments used in folding operations. Type I machines shall be similar to figure 1 and comply with the requirements of table I for folding length and capacity of machine ordered (see 6.2).



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Figure 1 - Type I, forming machine, bar folding.

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Table I - Type I forming machine, bar folding.

Maximum rated capacity in mild steel		Folding length (minimum)	Gaging range (minimum)	Shipping weight (plus or minus 10 percent)
Gage No.	Manufacturer's standard gage for sheet metal			
	Inch	Inches	Inch	Pounds
22	0.0299	20	1/8 to 1	150
22	.0299	30	1/8 to 1	250
20	.0359	20	1/8 to 1	170
20	.0359	30	1/8 to 1	240
20	.0359	36	1/8 to 1	390
20	.0359	42	1/8 to 1	500

3.3.1.2 Folding blade.- Folding blade shall be steel machine finished. A counterbalance arrangement shall be provided on machine 22 inches folding length and over to balance the folding bar, thereby facilitate ease of operation.

3.3.1.3 Clamping jaw.- Clamping jaw shall be actuated by the operating lever and shall grip and hold the sheet for all bending operations.

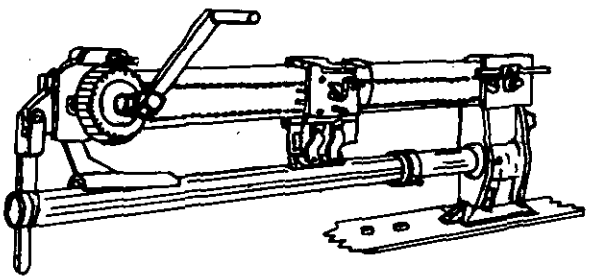
3.3.1.4 Stops.- Machines shall be provided with two positive stops of 45 degrees and 90 degrees and an adjustable stop to permit forming of any desired angle.

3.3.1.5 Gage.- An adjustable gage shall be provided, which shall be capable of controlling the width of the folds. The gage shall be provided with a scale graduated in increments of 1/16 inch. A thumb screw or other means shall be provided for locking the gage after the desired setting.

3.3.1.6 Wing.- Wing shall be of the adjustable type, which can be set for forming sharp bends and locks and can be adjusted back for open locks or rounding bends. Wing adjustment shall be provided by an adjusting wedge or other means. Wings shall be of good grade steel or cast iron of sufficient size and strength to perform all operations for which intended within the rated machine capacity.

3.3.1.7 Equipment.- Type I machines shall be furnished with all standard equipment normally furnished commercially with the type machine.

3.3.2 Type II forming machine.- Type II forming machines shall be of the hand operated bench type construction. They shall be suitable for closing inside or outside longitudinal seams. Machines shall consist of frame, base stand, horn, overarm carriage, rolls, shafts, bearings, gears and necessary levers, cranks and accessory equipment. Type II machines shall be similar to figure 2 and shall conform to the requirements of table II for the working length of machines specified (see 6.2).



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Figure 2 - Type II forming machine, grooving.

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Table II - Type II forming machine, grooving.

Maximum rated capacity in mild steel		Diameter of horn (maximum)	Working length (minimum)	Shipping weight (plus or minus 10 percent)
Gage No.	Manufacturer's standard gage for sheet metal			
	Inch	Inches	Inches	Pounds
22	0.0299	2	30	205
22	.0299	2	36	210
20	.0359	3	36	460
20	.0359	3-1/2	42	475

3.3.2.1 Frame and base.- Frame and base shall be of rugged construction and shall be of steel or cast iron. They shall be of such size and weight to substantially support the overarm, horn, carriage, and operating mechanisms and provide the alignment necessary for grooving operations.

3.3.2.2 Horn.- Horn shall be made of steel properly finished. It shall be provided with 3 longitudinal grooves for inside seams and a flat surface for outside seams. Horn shall be designed for rotating and locking in the desired position in the frame. A latch shall be provided for holding the horn in proper relation to the overarm in grooving operations. Latch shall be designed so it can be released for inserting work and shall be of sufficient strength and designed to perform all operations at rated capacity. The horn shall be provided with stops to fix position of work.

3.3.2.3 Overarm.- Overarm shall be made of cast iron or steel, rugged and so designed to support the carriage and latch arm and maintain the necessary alignment for all grooving operations within the rated machine capacity.

3.3.2.4 Carriage.- Carriage shall be designed for traveling on the overarm. It shall be driven by either an endless chain or rack and gear mechanism. Carriage shall hold the grooving and flattening rolls and shall be provided with pressure adjusting devices or other suitable mechanism for easily adjusting the pressure to conform to metal thicknesses. A device for reversing the direction of travel of the carriage shall be provided on machines of 20 gauge capacity or heavier.

3.3.2.5 Shafts and gears.- Shafts shall be of steel machined and fitted to suitable bearings. Gears shall be steel, machine cut, or cast iron. Gears shall be totally enclosed or guarded to protect the operator.

3.3.2.6 Operating cranks.- Machine shall be provided with all necessary operating cranks and levers for all operations within its rated capacity.

3.3.2.7 Equipment.- Each type II machine shall be furnished complete with the following equipment:

(a) Two each flattening rolls.
(b) Three each grooving rolls.

3.3.2.7.1 When required, extra rolls shall be furnished as specified (see 6.2).

3.4 Nameplate.- Unless otherwise specified (see 6.2), a corrosion resistant metal nameplate, permanently and legibly marked with the following information in accordance with MIL-STD-130 shall be securely attached to each machine:

Manufacturer's Name.
Manufacturer's Model Number.
Manufacturer's Serial Number.
Contract Number.
Date of Manufacture.
US.

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3.5 Identification plate.- When specified (see 6.2), all machines supplied shall carry a separate identification plate of corrosion resistant metal permanently and legibly marked with the following information. Figures shall be at least 3/16 of an inch in height.

Production equipment code number (to be supplied in contract).
Activity identification (to be inscribed by the receiving activity).

3.6 Handbook of instruction.- Unless otherwise specified (see 6.2), the contractor shall furnish two copies of manufacturer's standard handbook.

3.7 Workmanship.- Workmanship of the machine and accessories shall be of the quality prevailing among manufacturers normally producing sheet metal forming machines of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.-

4.2.1 Examination.- All machines shall be visually examined to determine compliance with the requirements of this specification. Any deviation from these requirements shall be cause for rejection.

4.2.2 Tests.- Each machine shall be tested at its maximum rated capacity in all operations for which designed and shall be completely tested with a minimum effort on the part of the operator. All forming operations shall be completely satisfactory without distortion to the workpiece.

4.2.3 Examination of preservation, packaging, packing and marking for shipment and storage.- Preservation, packaging, packing and marking for shipment and storage shall be examined to determine conformance with the documents referenced in section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing.- Unless otherwise specified (see 6.2), preservation, packaging, and packing shall be in accordance with MIL-P-18058. Level of preservation and packaging, and level of packing shall be as specified (see 6.2).

5.2 Marking.- Unless otherwise specified (see 6.2), marking shall conform to MIL-STD-129.

6. NOTES

6.1 Intended use.- The forming machines covered by this specification are intended for folding the edges of sheet metal and grooving or closing longitudinal seams in sheet metal cylinders after folding. These machines are for use in sheet-metal shops, aircraft maintenance shops and mobile shop trucks and repair units.

6.2 Ordering data.- Procurement documents shall specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Length and capacity (see tables I and II).
(d) Painting, if different (see 3.2.5).
(e) Extra rolls, if required (see 3.3.2.7.1).
(f) Nameplate, if different (see 3.4).
(g) Identification plate, if required (see 3.5).
(h) Handbook requirement, if different (see 3.6).
(i) Preservation, packaging, and packing, if different (see 5.1).

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(j) Level of preservation and packaging and level of packing required (see 5.1).
(k) Marking, if different (see 5.2).

6.3 Superseding data.- This specification covers the requirements of H folding machines and J grooving machines of MIL-M-1250.

Custodians:
Army - WC
Navy - SH
Air Force - 84

Preparing activity:
Navy - SH
(Project 3441-0007)

Review activity:
Navy - SH, YD
Air Force - 84

User activity:
Army - WC, MO
Navy - WP

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. *Comments and the return of this form will be appreciated.* Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

P 0700

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